Convolutional Architectures and CIFAR-100

The Uncs
Wesley Young, Jonathan Skelton, Brandon Riehl
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***Abstract*—CIFAR-100 contains 60,000 color images of size 32x32 across 100 classes, including 50,000 training and 10,000 test images. We train two CNN models from scratch and contrast their performance on CIFAR-100 to evaluate respective model design choices. Our models are a 2-block small CNN and a VGG-style deep CNN with batch norm. Each was trained with Adam at learning rates of 10^{-3} and 10^{-4} with and without dropout, resulting in 8 total configurations in the comparison. The main result was that the deep CNN was the better model (learning rate 10^{-4}, p = 0.3), with 62.4% top-1 (95% CI on the error: [0.366, 0.385]) and 88.1% top-5. The best small model reached 44.7% top-1 and 74.4% top-5. Dropout reduced the Deep model's error by 0.034 at the lower learning rate, while the combination of a higher learning rate and dropout substantially reduced accuracy. The deeper architecture with batch normalization achieved the best overall performance. Integrated Gradients in the appendix show the remaining errors come from visually similar classes.**

Introduction

CIFAR-100 presents a unique challenge in developing an accurate model because it has 100 classes with only 500 training images per-class. Many classes are difficult to differentiate (e.g., boy/man, maple/oak). The goal of our project is to measure how architecture depth, learning rate, and dropout each affect generalization. We use confidence intervals alongside raw accuracy to evaluate model performance. From our experiment, we observed our deep model to outperform the smaller model by 17.7 points of top-1 accuracy reaching 62.4% top-1 and 88.1% top-5.

I. Problem and Methods

The input to the CNN are 32 x 32 pixel images with 3 color channels in batches of 64. Each image is classified into one of one hundred classes (e.g. wolf, house, or bicycle). Each class also belongs to a superclass such as large carnivores or large man-made outdoor things. The models were trained and tested on the classes but superclass accuracy is also reported by mapping each predicted class to its parent superclass. The input image pixel values are scaled to the 0-1 range in the input layer by dividing by the maximum pixel value of 255. Class labels were converted into one-hot vectors and categorical cross-entropy was used as the loss function. All networks used minor data augmentation during training. The input images were randomly flipped horizontally and randomly translated horizontally and vertically by up to 10% of the image size (reflect fill). Augmentation was applied only to training images.

Two CNN architectures were evaluated. The Small model (Table S1) has two convolution pooling blocks with 32 and 64 filters, followed by a dense layer of 256 nodes. The Deep model (Table I) contains three blocks with two convolutions and batch normalization per block, with 64, 128, and 256 filters, followed by a dense layer of 512 nodes. Both models use ReLU, dropout when $p = 0.3$, and a 100-class softmax output.

All models were trained using Adam and L2 weight regularization ($\lambda = 5 \times 10^{-4}$), following VGG [2]. L2 strength was fixed to isolate the learning rate and dropout effects. Both architectures were tested at learning rates 10^{-3} (Adam default) and 10^{-4}, and dropout rates $p = 0$ and $p = 0.3$, for eight configurations. The lower learning rate tested more stable convergence, while the dropout rate provided moderate regularization.

The 50,000 image training set was divided into 90% training and 10% validation, which is the same as Fashion MNIST. The validation set was used for early stopping with a patience parameter of 10 epochs (above the required minimum of 5, because validation accuracy fluctuated between epochs) and restored weights for the best model. Training and validation loss and accuracy vs. epoch are plotted in Fig. S1. Final top-1 and top-5 accuracies on the test set are reported in this paper. We calculated 95% confidence intervals using the normal approximation for individual generalization error rates and paired per-image error differences for model comparisons ($n = 10{,}000$).

II. Experimental Results

The best Small CNN achieved 44.7% top-1 and 74.4% top-5 test accuracy at learning rate 10^{-4} and dropout $p = 0.3$. The best Deep CNN achieved 62.4% top-1 and 88.1% top-5 with the same settings. Its generalization error was 0.376 (95% CI [0.366, 0.385]). The Deep model reduced the error by 0.177 compared with Small (95% CI [0.167, 0.187]). Within the Deep architecture, dropout reduced error by 0.034 (95% CI [0.02, 0.04]). All eight test results are reported in Table S2.

The best model had the highest prediction accuracy on the Trees superclass (Fig. 1). For conciseness, the normalized confusion matrix for the superclasses is presented here. See the Appendix for the normalized 100 class confusion matrix (Fig. S3). The model had the most confusion on similar superclasses such as confusing aquatic mammals and fish or confusing large carnivores with medium mammals.

Motorcycle was the most accurate class (93%), whale the median (65%), and seal the least accurate (17%, Table S3). Integrated Gradients (IG) [3] were compared across these three classes. For each, we selected the highest confidence correct prediction, lowest confidence correct prediction, and highest

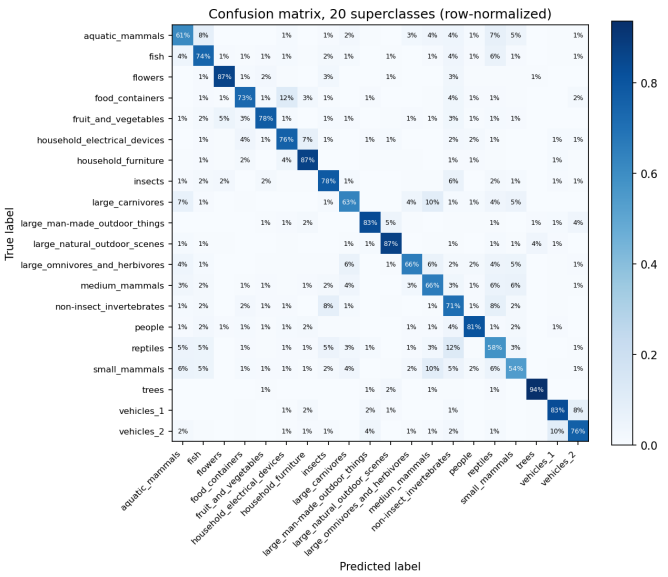


Fig. 1. Normalized confusion matrix for superclasses

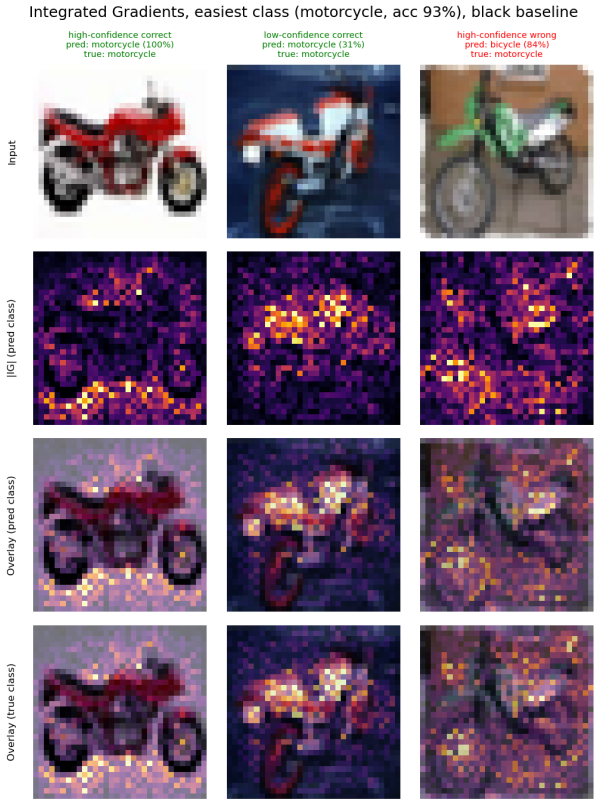


Fig. 2. Integrated gradients for the easiest class for (left) high confidence correct, (middle) low confidence correct, and (right) high confidence incorrect

confidence incorrect prediction, using maximum softmax probability as confidence. In Fig. 2, attribution for correct motorcycle predictions emphasized the motorcycle's surroundings or body, whereas the incorrect prediction emphasized wheels and fairing and was classified as bicycle. Similar border emphasis appears for whale and seal (Figs. S4-S5).

The most confident misclassifications and selected IG maps appear in Figs. S6-S7. In one example, a forest image was predicted as mountain, with IG emphasizing the mountainous background. Another was a clock predicted as telephone, had less interpretable attributions, potentially influenced by the clock's wedge shape.

III. Discussion

Of the eight configurations, the VGG-inspired Deep CNN performed best. At dropout $p = 0.3$, lowering the learning rate from 10^{-3} to 10^{-4} raised top-1 accuracy from 30.5% to 62.4%. Without dropout, the improvement was smaller (57.1% to 59.0%). The high learning rate Deep run with dropout struggled to converge, with training accuracy below 25% (Fig. S1). This suggests difficulty with optimization rather than overfitting in that configuration. At 10^{-4}, dropout improved top-1 accuracy from 59.0% to 62.4% and top-5 accuracy from 85.0% to 88.1%. These results suggest learning rate and dropout should be tuned together. Further experiments could examine intermediate dropout rates, architecture widths, and augmentation.

Fig. S1 shows overfitting in the Deep CNN without dropout at 10^{-4}. Training accuracy reached about 84% while validation accuracy plateaued near 58%. Validation loss stopped improving after about epoch 35 even as training loss continued to decline, indicating that further fitting did not improve generalization. With dropout $p = 0.3$, training accuracy fell to 66% while test accuracy increased from 59.0% to 62.4%. Small models showed smaller training-validation gaps (at most 6 points), suggesting limited capacity. At 10^{-4}, dropout changed the Small model's top-1 test accuracy only slightly, from 44.5% to 44.7%. At 10^{-3}, adding dropout reduced its accuracy from 44.4% to 40.0%.

Strengths include controlled comparisons, early stopping with restored weights, and reporting confidence intervals, top-5 accuracy, normalized confusion matrices, and IG maps. With 100 classes, top-5 accuracy indicates whether the correct class is among the five highest-scoring predictions. Limitations include one training run per configuration, only two tested values per hyperparameter, and fixed L2 strength. Since the Deep CNN introduced batch normalization alongside greater depth, their effects can't be isolated.

IV. Conclusion

The 3-block VGG-style CNN with batch normalization outperformed the Small CNN across the eight tested configurations. Learning rate and dropout interacted a great deal: the Deep model with dropout at 10^{-3} trained poorly, while dropout at 10^{-4} reduced overfitting and improved test accuracy. The Small model seemed more capacity limited. Errors often involved visually related classes, and Integrated Gradients highlighted some plausible sources of confusion, such as mountains in a forest image.

REFERENCES

[1] A. Krizhevsky, "Learning Multiple Layers of Features from Tiny Images," Tech. Rep., Univ. of Toronto, 2009.

[2] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," Apr. 10, 2015, arXiv:1409.1556. doi: 10.48550/arXiv.1409.1556.

[3] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic Attribution for Deep Networks," in Proc. 34th Int. Conf. Mach. Learn. (ICML), 2017, pp. 3319–3328.

APPENDIX

TABLE I
SUMMARY OF DEEP CNN

Stage	**Configuration**
Input	RGB image, 32 x 32 x 3 (NHWC)
Scaling	Rescaling(1/255), from 0-255 pixels to 0-1
Augmentation	Horizontal flip; +/-10% H/W translation (reflect fill; training only)
Conv block 1	2 x [Conv2D(64, 3x3, stride 1, same, bias=False)-> BatchNorm->ReLU]->MaxPool2D(2x2, stride=2)-> Dropout(d)
Conv block 2	2 x [Conv2D(128, 3x3, stride 1, same, bias=False)-> BatchNorm->ReLU]->MaxPool2D(2x2, stride=2) -> Dropout(d)
Conv block 3	2 x [Conv2D(256, 3x3, stride 1, same, bias=False) -> BatchNorm->ReLU]->MaxPool2D(2x2, stride=2)-> Dropout(d)
Classifier	Flatten -> Dense(512, ReLU) -> Dropout(d)
Output	Dense(100, Softmax)
Regularization	L2 = 5e-04 on Conv2D and hidden Dense kernels; d in {0.0, 0.3}; omit Dropout if d=0
Defaults	Glorot-uniform kernels; zero-initialized biases; BatchNorm momentum=0.99, epsilon=0.001

TABLE S1
SUMMARY OF SMALL CNN

Stage	**Configuration**
Input	RGB image, 32 x 32 x 3 (NHWC)
Scaling	Rescaling(1/255), from 0-255 pixels to 0-1
Augmentation	Horizontal flip; +/-10% H/W translation (reflect fill; training only)
Conv block 1	Conv2D(32, 3x3, stride 1, same, bias=True) -> ReLU-> MaxPool2D(2x2, stride 2) -> Dropout(d)
Conv block 2	Conv2D(64, 3x3, stride 1, same, bias=True) -> ReLU-> MaxPool2D(2x2, stride 2) -> Dropout(d)
Classifier	Flatten -> Dense(256, ReLU) -> Dropout(d)
Output	Dense(100, Softmax)
Regularization	L2 = 5e-04 on Conv2D and hidden Dense kernels; d in {0.0, 0.3}; omit Dropout if d=0
Defaults	Glorot-uniform kernels; zero-initialized biases

TABLE S2
SUMMARY OF TEST ACCURACY FOR EACH CONDITION

Arch.	**Learning Rate**	**Dropout**	**L2**	**Params**	**Best Epoch**	**Test Accuracy**	**Test Top5 Accuracy**
Deep	0.0001	0.3	0.0005	3297060	68	0.62449	0.88099
Deep	0.0001	0	0.0005	3297060	36	0.59039	0.84950
Deep	0.001	0	0.0005	3297060	41	0.57090	0.84399
Deep	0.001	0.3	0.0005	3297060	23	0.30540	0.64880
Small	0.0001	0.3	0.0005	1093924	135	0.44749	0.74419
Small	0.0001	0	0.0005	1093924	91	0.44519	0.73299
Small	0.001	0	0.0005	1093924	33	0.44359	0.74299
Small	0.001	0.3	0.0005	1093924	31	0.40029	0.71289

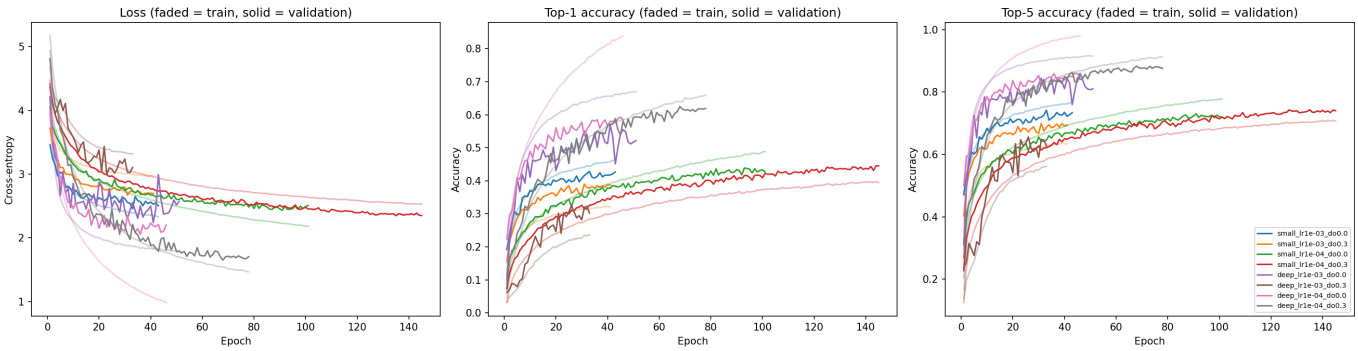


Fig. S1. Loss and Accuracy vs. Training Epoch

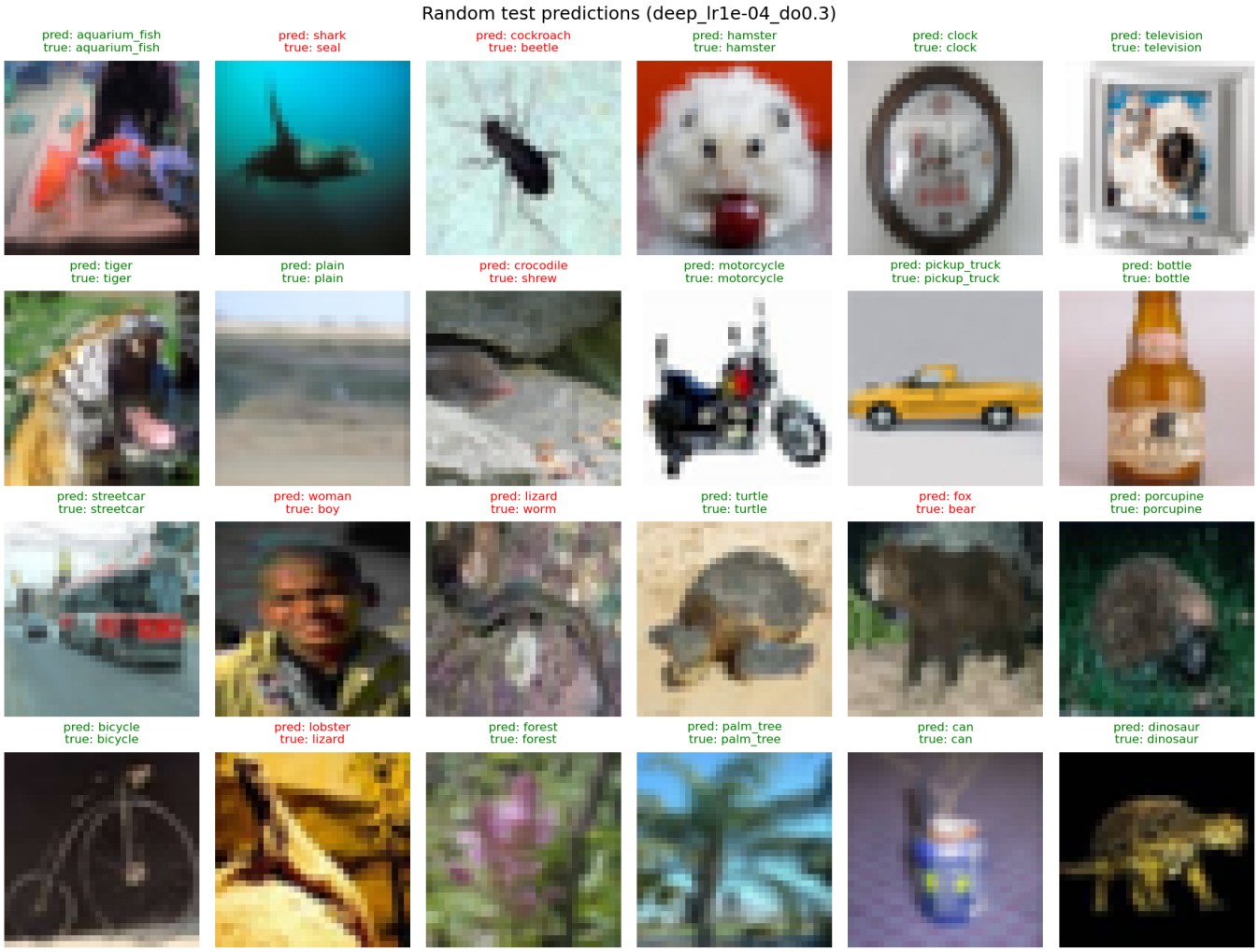


Fig. S2. Random accurate and inaccurate predictions from the test set

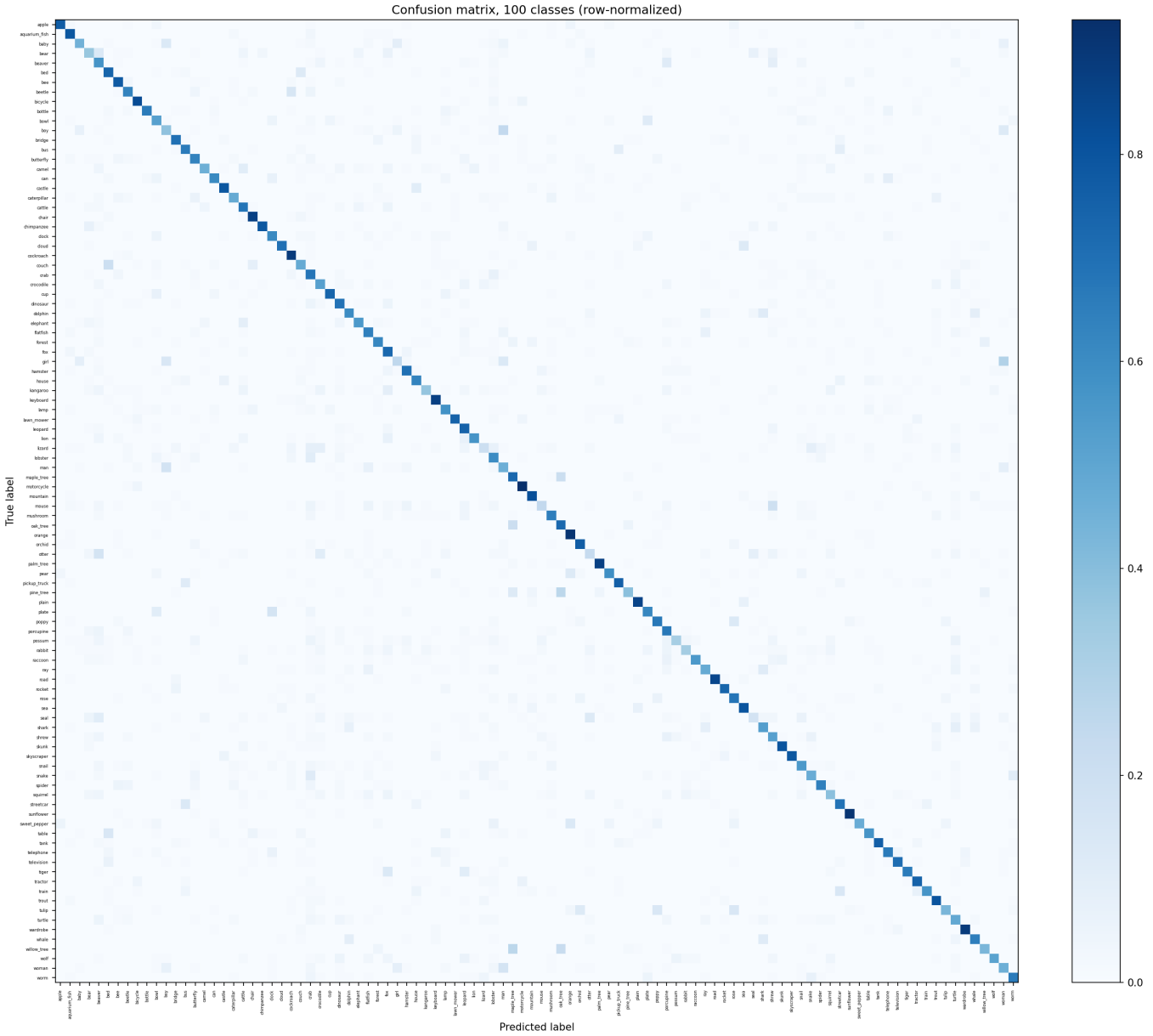


Fig. S3. Normalized confusion matrix for all 100 classes

TABLE S3: Summary of Class, Accuracy, and Most Confused Class

Class	Superclass	Accuracy	Most Confused With:	Times Confused
apple	fruit_and_vegetables	0.77	orange	4
aquarium_fish	fish	0.79	crab	3
baby	people	0.45	boy	13
bear	large_carnivores	0.36	beaver	13
beaver	aquatic_mammals	0.57	porcupine	11
bed	household_furniture	0.74	couch	12
bee	insects	0.78	beetle	3
beetle	insects	0.63	cockroach	15
bicycle	vehicles_1	0.83	motorcycle	5
bottle	food_containers	0.66	telephone	5
bowl	food_containers	0.52	plate	11
boy	people	0.38	man	24
bridge	large_man-made_outdoor_things	0.7	forest	4
bus	vehicles_1	0.66	pickup_truck	9
butterfly	insects	0.63	leopard	4
camel	large_omnivores_and_herbivores	0.46	cattle	10
can	food_containers	0.61	telephone	9
castle	large_man-made_outdoor_things	0.79	house	10
caterpillar	insects	0.48	butterfly	6
cattle	large_omnivores_and_herbivores	0.68	fox	5
chair	household_furniture	0.87	couch	3
chimpanzee	large_omnivores_and_herbivores	0.79	bear	7
clock	household_electrical_devices	0.61	bowl	5
cloud	large_natural_outdoor_scenes	0.7	sea	12
cockroach	insects	0.89	beetle	2
couch	household_furniture	0.48	bed	20
crab	non-insect_invertebrates	0.68	lobster	7
crocodile	reptiles	0.52	leopard	5
cup	food_containers	0.74	bowl	7
dinosaur	reptiles	0.69	crab	5
dolphin	aquatic_mammals	0.6	shark	11
elephant	large_omnivores_and_herbivores	0.55	cattle	9
flatfish	fish	0.64	ray	7
forest	large_natural_outdoor_scenes	0.62	willow_tree	7
fox	medium_mammals	0.73	hamster	4
girl	people	0.25	woman	29
hamster	small_mammals	0.7	fox	4
house	large_man-made_outdoor_things	0.63	castle	6
kangaroo	large_omnivores_and_herbivores	0.37	fox	9
keyboard	household_electrical_devices	0.87	couch	2
lamp	household_electrical_devices	0.58	chair	4
lawn_mower	vehicles_2	0.75	motorcycle	4
leopard	large_carnivores	0.75	shrew	4
lion	large_carnivores	0.57	fox	7
lizard	reptiles	0.2	crab	8
lobster	non-insect_invertebrates	0.6	crab	10
man	people	0.46	boy	17
maple_tree	trees	0.73	oak_tree	21
motorcycle	vehicles_1	0.93	bicycle	4
mountain	large_natural_outdoor_scenes	0.82	whale	3
mouse	small_mammals	0.24	shrew	19
mushroom	fruit_and_vegetables	0.65	leopard	4
oak_tree	trees	0.73	maple_tree	21
orange	fruit_and_vegetables	0.91	aquarium_fish	1
orchid	flowers	0.77	butterfly	3
otter	aquatic_mammals	0.23	beaver	16
palm_tree	trees	0.87	maple_tree	2
pear	fruit_and_vegetables	0.59	orange	12
pickup_truck	vehicles_1	0.76	bus	11
pine_tree	trees	0.39	oak_tree	26
plain	large_natural_outdoor_scenes	0.86	sea	7

Table S3 (continued)

Class	Superclass	Accuracy	Most Confused With:	Times Confused
plate	food_containers	0.62	clock	16
poppy	flowers	0.67	rose	13
porcupine	medium_mammals	0.67	beaver	5
possum	medium_mammals	0.32	turtle	6
rabbit	small_mammals	0.33	porcupine	5
raccoon	medium_mammals	0.55	porcupine	6
ray	fish	0.48	shark	11
road	large_man-made_outdoor_things	0.85	bridge	3
rocket	vehicles_2	0.74	bridge	5
rose	flowers	0.68	poppy	6
sea	large_natural_outdoor_scenes	0.8	plain	7
seal	aquatic_mammals	0.17	otter	12
shark	fish	0.53	ray	8
shrew	small_mammals	0.52	porcupine	7
skunk	medium_mammals	0.8	cattle	3
skyscraper	large_man-made_outdoor_things	0.79	castle	5
snail	non-insect_invertebrates	0.55	lizard	4
snake	reptiles	0.47	crab	11
spider	non-insect_invertebrates	0.64	bee	5
squirrel	small_mammals	0.39	fox	5
streetcar	vehicles_2	0.69	bus	13
sunflower	flowers	0.91	rose	2
sweet_pepper	fruit_and_vegetables	0.47	orange	13
table	household_furniture	0.57	bed	14
tank	vehicles_2	0.76	pickup_truck	3
telephone	household_electrical_devices	0.66	clock	6
television	household_electrical_devices	0.71	bed	6
tiger	large_carnivores	0.66	fox	13
tractor	vehicles_2	0.75	motorcycle	6
train	vehicles_1	0.57	streetcar	14
trout	fish	0.8	lizard	3
tulip	flowers	0.43	poppy	14
turtle	reptiles	0.49	beaver	5
wardrobe	household_furniture	0.89	television	3
whale	aquatic_mammals	0.65	shark	11
willow_tree	trees	0.43	maple_tree	26
wolf	large_carnivores	0.51	leopard	6
woman	people	0.46	man	16
worm	non-insect_invertebrates	0.66	snake	7

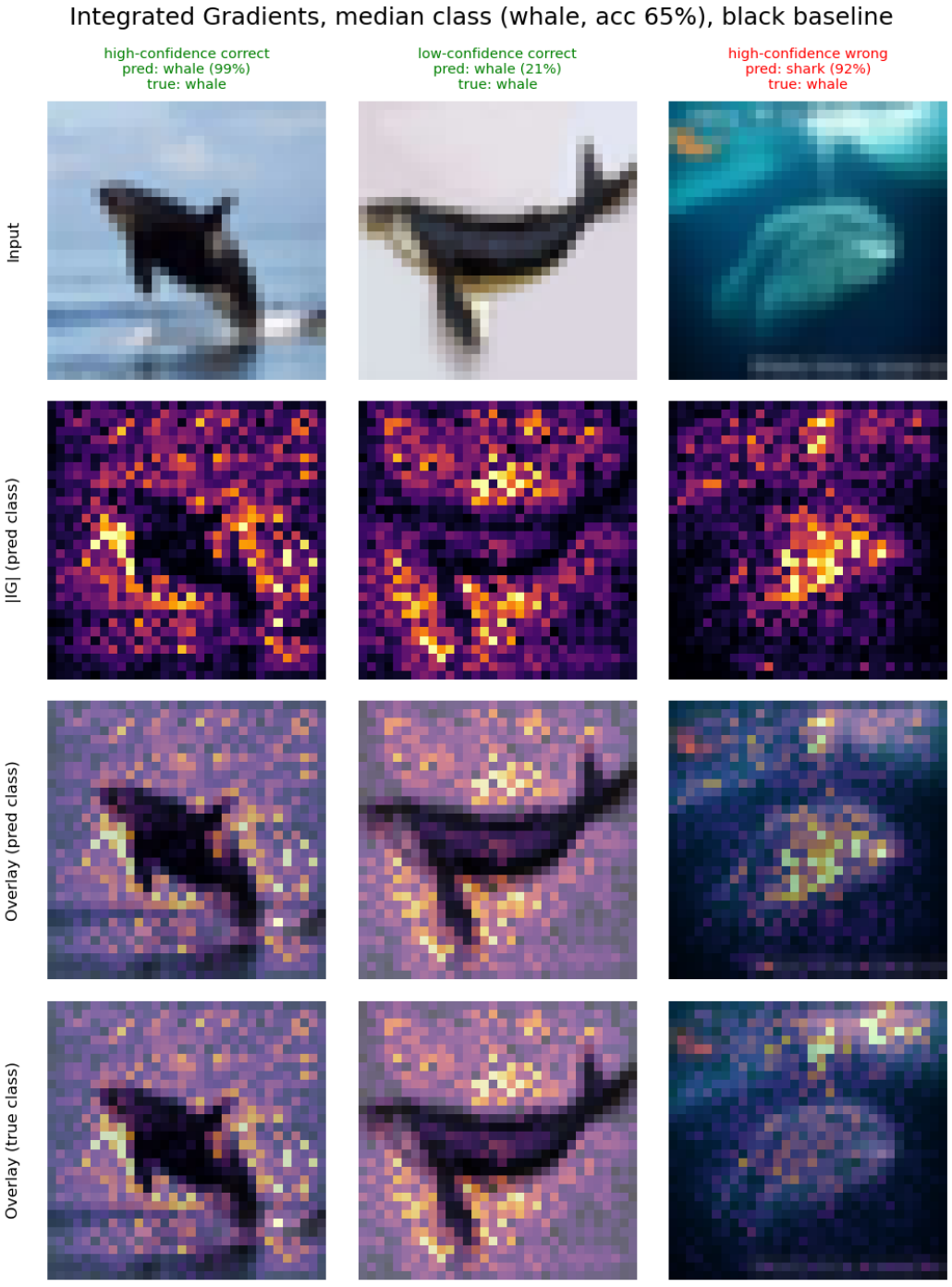


Fig. S4. Integrated gradients for the median class for (left) high confidence correct, (middle) low confidence correct, and (right) high confidence incorrect

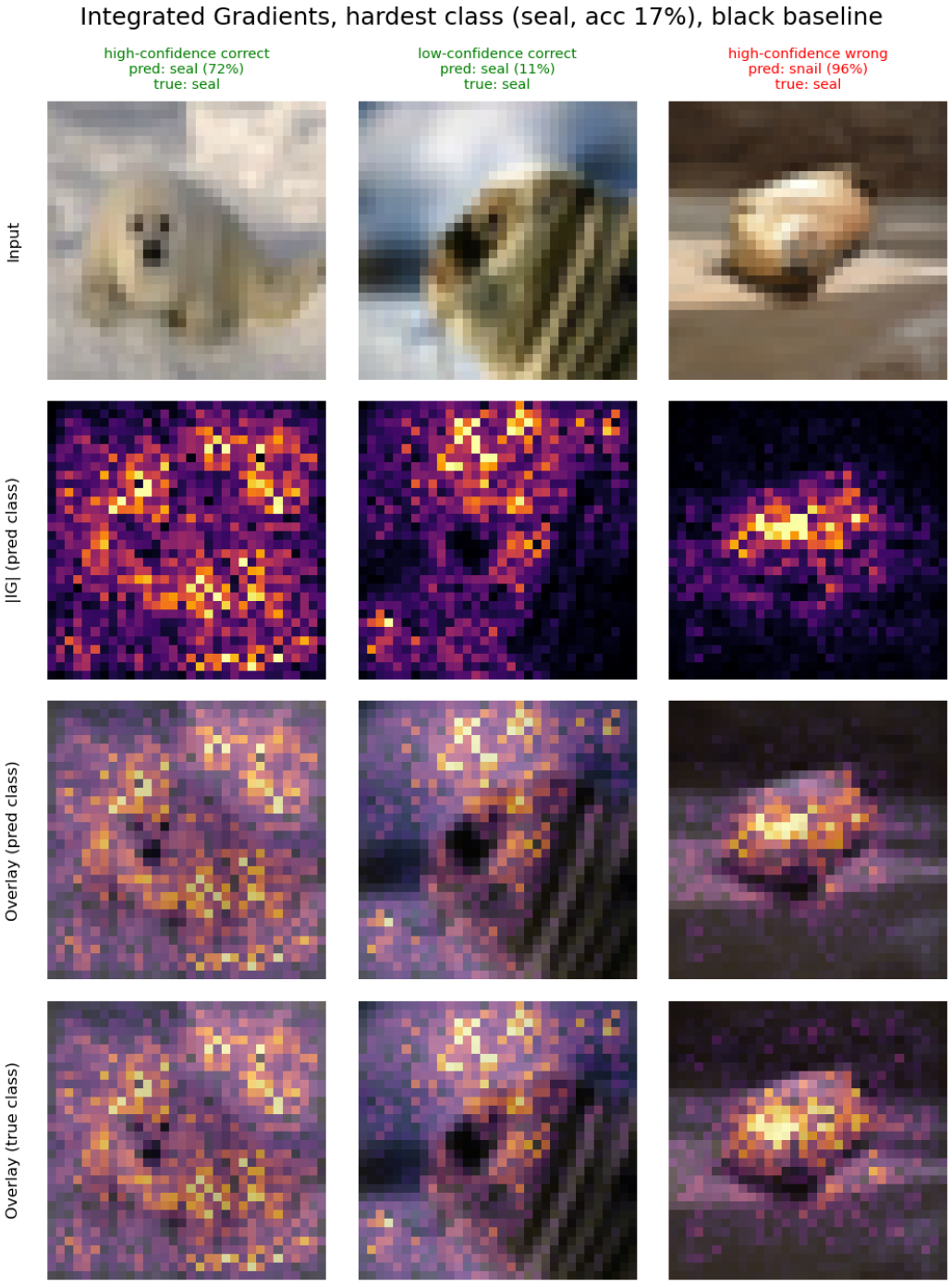


Fig. S5. Integrated gradients for the hardest class for (left) high confidence correct, (middle) low confidence correct, and (right) high confidence incorrect

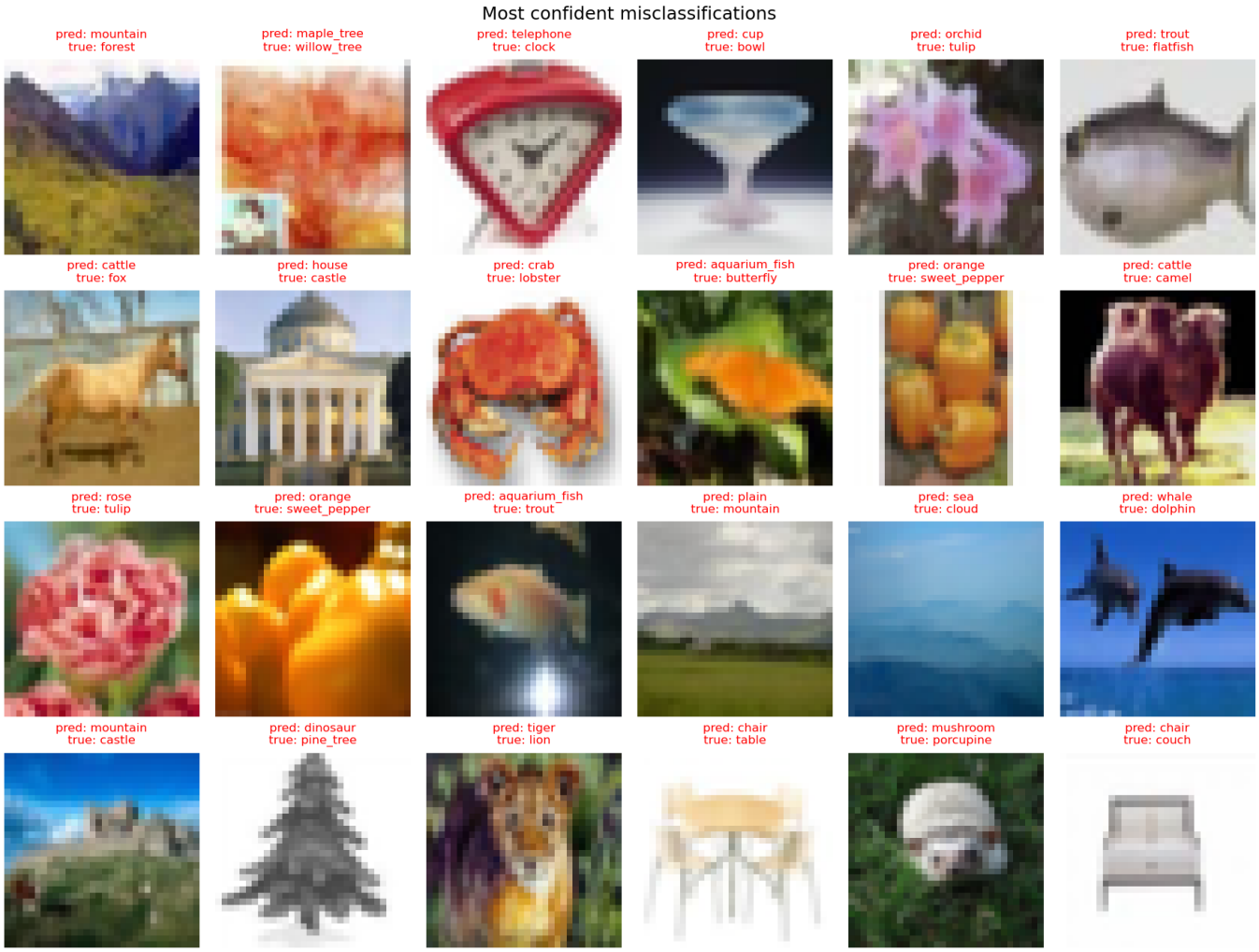


Fig. S6. Most confident misclassifications

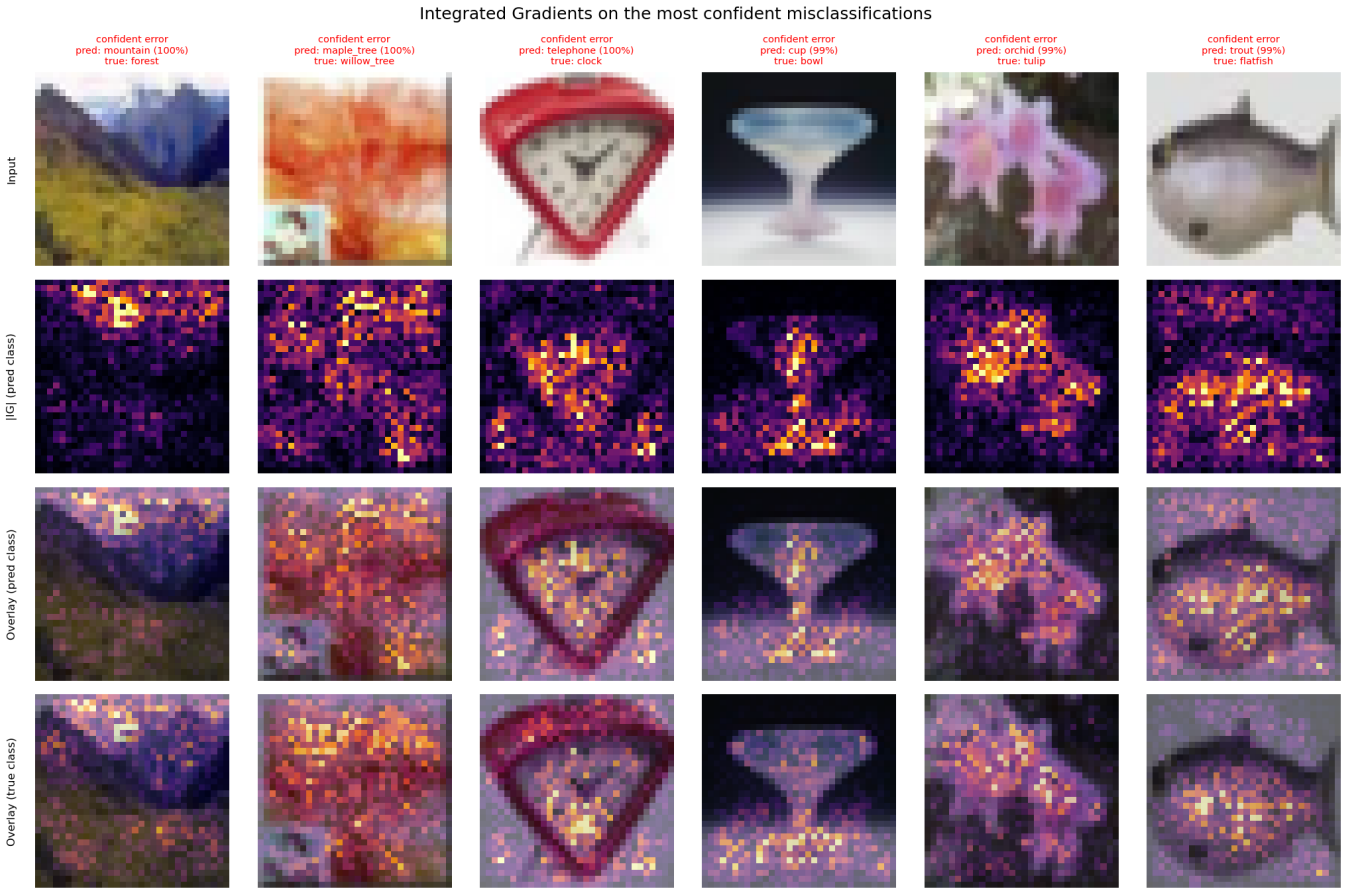


Fig. S7. Integrated gradients for a subset of the most confident misclassifications